NAI-WEN CHI

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RESEARCH INTERESTS

Targeted nucleic-acid delivery; intracellular trafficking and productive gene expression; in vivo immune-cell engineering; biomaterials; human-relevant translational models.

EDUCATION

M.S., Biomedical Engineering | National Tsing Hua University, Taiwan | Feb 2015 - Jan 2017

GPA: 4.05/4.30. Thesis research: nitric-oxide-releasing microparticles for osteoporosis therapy.

• Developed phase-change delivery systems; performed in vitro assays and studies using an ovariectomized rat model. Research contributed to an Advanced Materials publication and patent filings.

B.S., Agronomy | National Chung Hsing University, Taiwan | Sep 2008 - Jun 2012

GPA: 3.25/4.00.

RESEARCH & TRANSLATIONAL EXPERIENCE

R&D & Regulatory Scientist (Trainee) | Coastar Therapeutics, California | Dec 2025 - Present

• Conduct gap analysis and support FDA development strategy for in vivo CAR-T therapeutics; contributed to preparation for an FDA INTERACT meeting completed in June 2026.

Research Assistant | Academia Sinica, Taiwan | Apr 2025 - Nov 2025

• Assisted cancer-targeted drug-delivery research under Dr. Yu-Jung Lin, designing and evaluating carrier systems in cell-based and mouse models.

R&D Engineer | BioGend Therapeutics, Taiwan | Jan 2018 - Mar 2021

• Contributed to the RevoCart cartilage-repair implant across design, preclinical testing, clinical support, and regulatory development. The platform achieved TFDA registration and entered clinical use in Taiwan.

• Supported PLGA/TCP scaffold fabrication, Instron mechanical testing, mini-pig implantation studies, and usability testing with 15 international surgeons.

• Developed tyramine-alginate hydrogel systems for cartilage adhesion and regeneration; co-authored a publication in International Journal of Biological Macromolecules.

• Provided technical support during human clinical surgical cases, including equipment setup and instrument sterilization coordination.

R&D Engineer | Exactech Taiwan Ltd. | May 2017 - Dec 2017

• Supported R&D projects and clinical trials for Class III devices. Exactech Taiwan was acquired by BioGend in January 2018.

PUBLICATIONS

Lin YJ, Chen CC, **Chi NW**, Nguyen T, Lu HY, Nguyen D, Lai PL, Sung HW. In Situ Self-Assembling Micellar Depots that Can Actively Trap and Passively Release NO with Long-Lasting Activity to Reverse Osteoporosis. Advanced Materials. 2018;30(22):1705605. doi:10.1002/adma.201705605.

Wong CC, Lu CX, Cho EC, Lee PW, **Chi NW**, Lin PY, Jheng PR, Chen HL, Mansel BW, Chen YM, Chen CH, Chuang EY. Calcium peroxide aids tyramine-alginate gel to crosslink with tyrosinase for efficient cartilage repair. International Journal of Biological Macromolecules. 2022;208:299-313. doi:10.1016/j.ijbiomac.2022.03.044.

REGULATORY & PROJECT EXPERIENCE

Senior Regulatory Affairs Specialist | Winnoz Technology, Taiwan | May 2022 - Jul 2023

• Led TFDA Class I device license applications and amendments, obtaining two licenses. Prepared documentation for EU IVDR Class B and U.S. FDA Class II IVD submissions.

• Supported ISO 13485 and TFDA quality-management-system audits.

Regulatory Affairs Specialist / Project Manager | BioGend Therapeutics | Apr 2021 - May 2022

• Managed TFDA Class I-III medical-device license applications and secured seven approvals.

• Independently led a TFDA Class II PRP-device application, coordinating more than 30 preclinical studies through approval.

• Prepared documentation for planned EU, U.S., China, and Vietnam submission strategies; these submissions were not filed following company strategy decisions.

• Supported design controls, ISO 13485 audits, and TFDA QMS audits.

Chief Executive Officer & Project Manager | Wen Fu AI Co., Taiwan | Jul 2023 - Mar 2024

• Founded and led a small technology venture, managing online-course and AI-assisted website projects, including scope, schedules, and client communication.

Career Transition & Graduate Study Preparation | Taiwan | Mar 2024 - Mar 2025

Prepared doctoral applications and completed English proficiency testing.

TECHNICAL & ANALYTICAL SKILLS

Delivery systems: nanoparticle and microparticle fabrication; microfluidic O/W emulsions; phase-change and thermoresponsive materials; HPLC-based drug-encapsulation analysis.

Cell and animal models: primary dorsal-root-ganglion neuron culture; MC3T3-E1 osteoblast culture; mouse breast-cancer xenografts; ovariectomized rat osteoporosis model.

Biomedical assays: WST-1 viability; alkaline-phosphatase activity; Alizarin Red S mineralization; H&E histology; serum ALP and osteocalcin.

Characterization: differential scanning calorimetry; micro-CT bone-density analysis; confocal microscopy; nitric-oxide release kinetics; ultrasound imaging.

Regulatory: FDA INTERACT preparation and IND strategy support; TFDA Class I-III; EU MDR/IVDR documentation; ISO 13485; design controls.

SELECTED PATENT FILINGS

Named co-inventor on temperature-responsive drug-delivery and cartilage-repair-device invention families. The statuses below reflect the August 2026 source CV and have not been rechecked for subsequent changes.

• Temperature-Responsive Composition, Method for Fabricating, and Use Thereof: Taiwan invention patent TWI606829B, granted 2017; later lapsed for non-payment of annuities. U.S. published application US20180161276A1, abandoned.

• Cartilage Filling System and Surgical Instrument Set: Taiwan utility model TWM594960U; China utility model CN212730081U.

• Cartilage Volumetric Device and Surgical Instrument Set: Taiwan utility model TWM594437U; China utility model CN212729944U.

• Dry-Type Heating Device: Taiwan utility model TWM590460U; China utility model CN211454393U.

LANGUAGES

Mandarin: native. English: professional working proficiency.